Session 3 (DESIGN): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · *#ClearDataClearImpact* · Week 3 · Saturday 14 November 2026 · online, 19:30–22:30

Name: _____ **Department:** _____

Type your answers in the Word file, or print the PDF. No calculator is needed.

Part	Task	When
A	Name the design of 6 short abstracts	End of section 3.1
B	Choose a design for 5 planning questions, then for your own question	End of section 3.2
C	Proposal builder 3: draft sections 2.1 to 2.3 of your proposal	End of the session, finish at home

Each part also has **extra practice** on other topics, for self-study after the session.

Our running question: in adults with septic shock, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?

Abbreviations: ICU = intensive care unit, RCT = randomised controlled trial.

Part A: Name that design

Task. Read each abstract alone. Write its design (choose from the list below) and the words in the abstract that show it. Be ready to share your answer when the instructor asks.

Three questions name the design

1. **Did the investigator assign the exposure or treatment?** If yes, **at random?** Yes: RCT. No: non-randomised intervention study.
2. If not assigned: **is there a comparison group?** No: descriptive (case report or case series).
3. If there is: **which came first?** Exposure first, then outcome: cohort. Outcome first, then look back: case-control. Both at the same moment: cross-sectional.

> **Choose from:** case report or case series · cross-sectional · case-control · cohort (prospective or retrospective) · non-randomised intervention study · RCT · systematic review and meta-analysis

Abstract 1. Two hundred adults with septic shock were randomly allocated by a central computer to a restrictive fluid strategy or to usual care. Death by day 28 was recorded for every patient.

Design: _____ Words that show it: _____

Abstract 2. On one day in March, two doctors reviewed all 64 patients with sepsis in four ICUs. For each patient they recorded the fluid balance that day and whether the patient had acute kidney injury that day.

Design: _____ Words that show it: _____

Abstract 3. Forty ICU patients with sepsis who needed new dialysis were compared with 80 ICU patients with sepsis from the same units who did not. The investigators then looked back in the charts at each patient's fluid balance on day 3.

Design: _____ Words that show it: _____

Abstract 4. Using electronic records from 2019 to 2022, the investigators identified 900 patients with septic shock and divided them by whether their fluid balance on day 3 was positive or negative. They then compared how many patients in each group died by day 28.

Design: _____ Words that show it: _____

Abstract 5. Our ICU introduced a fluid-restriction protocol for all patients with septic shock on 1 January 2024. Mortality in the six months before was compared with mortality in the six months after.

Design: _____ Words that show it: _____

Abstract 6. We searched three databases for randomised trials comparing restrictive with liberal fluid strategies in adults with septic shock, assessed each trial's risk of bias, and pooled 28-day mortality across nine trials.

Design: _____ Words that show it: _____

Extra practice (self-study): other topics

Same task, nine abstracts from other wards. The answers are in the solutions sheet, shared after the session.

E1. Two hundred and forty adults on mechanical ventilation were randomly allocated by a central computer to oral care with chlorhexidine twice daily or to standard oral care. Ventilator-associated pneumonia was recorded over 28 days by an assessor who did not know the group.

E2. On one day in March, two trained nurses examined all 412 inpatients of a hospital for pressure injuries and recorded the type of mattress each patient was lying on. Pressure injuries were present in 9% of patients on standard mattresses and 4% on pressure-relieving mattresses.

E3. Forty ICU patients who developed candidaemia (Candida in the blood) were compared with 80 ICU patients from the same units who did not, matched on age and month of admission. The investigators then searched the records of both groups for parenteral nutrition in the 14 days before the culture date.

E4. All 600 adults admitted with acute ischaemic stroke who received thrombolysis were classified on arrival by door-to-needle time (under 60 minutes, or 60 minutes and over). Each patient was then followed up to record functional independence at 90 days.

E5. Using electronic records from 2019 to 2022, the investigators identified 1,150 ICU patients and divided them into those who did and did not receive a continuous benzodiazepine infusion for 3 days or more. They then compared how many patients in each group had delirium recorded later in the ICU stay.

E6. We describe five patients who developed severe hypernatraemia within a week of a new fluid protocol being introduced on our unit, with their laboratory results and clinical course.

E7. Central-line infection rates were compared for the six months before and the six months after a unit-wide insertion-bundle education programme, which the research team introduced on all beds on the same day.

E8. Twelve hospital wards were randomly allocated, ward by ward, to a new hand-hygiene feedback programme or to usual practice. Hospital-acquired infections were counted in all patients on those wards for six months.

E9. All 300 inpatients are checked today for a urinary catheter and for urinary infection.

Part B: Choose the design

Task. For each planning question, write the best design you could actually carry out, one reason for it, and its main weakness. Be ready to share your answer when the instructor asks.

> **Choose from:** any design in Part A, or one of the designs from section 3.2: non-inferiority RCT · cluster RCT · interrupted time series · qualitative interviews or focus groups · pilot or feasibility study

#	Planning question (all on our fluid question)	Design	Reason	Main weakness
1	A restrictive strategy is cheaper and uses less fluid. We only need to show that 28-day mortality is not worse than with usual care.			
2	A nurse-led fluid-balance charting protocol will be introduced in whole ICUs. Nurses in one ICU cannot use it for only half their patients.			
3	Our hospital's sepsis bundle, which includes a limit on maintenance fluids, started in 2023. We have monthly sepsis mortality for 3 years before and 2 years after. Did mortality fall?			
4	Why do night-shift nurses find it hard to record urine output?			
5	Before a large trial: can our two ICUs recruit enough patients and deliver restrictive fluids as planned?			

Your own question. Write your research question, the design you will use, and what you will be allowed to say at the end: "caused" or only "associated with".

My question: _____

My design: _____ I may say: "caused" / "associated with" (circle one)

Extra practice (self-study): other topics

#	Question	Design	Reason
E1	How many inpatients have a pressure injury today, and is it more common on standard mattresses?		
E2	Does daily chlorhexidine bathing reduce bloodstream infections? It will be introduced ward by ward.		
E3	What are the risk factors for a rare event: Stevens-Johnson syndrome in hospital patients?		
E4	Why do nurses not report near-miss medication errors?		

Part C: Proposal builder 3 (design, setting and population)

Task. Draft sections 2.1 to 2.3 of your proposal for your own question. First watch the instructor fill them in for the running example. Finish at home, then copy your answers into Part 2 of the proposal template (`Research_Proposal_Template.docx`). Sections 2.4 to 2.6 (exposure, outcomes, bias) follow in Session 4.

My research question (from Part 1 of my proposal):

_____

Section	What to write	My answer
2.1 Study design and justification	Name the design. In 3 to 5 sentences, say why it fits your question, and why the ideal design is or is not feasible.	
2.2 Setting	Where (type of unit, how many sites), when (study period), and the source of patients or records. You do not need to name your hospital.	
2.3 Study population	**Inclusion criteria** (3 to 5): who is eligible. **Exclusion criteria** (3 to 5): who is removed, and why. An exclusion is not just an inclusion turned negative.	

Sketch your study on a timeline. Write when the exposure is measured and when the outcome is measured:

Start: exposure measured	Follow-up	End: outcome measured

Self-check before you finish

- ☐ The design matches the type of question.
- ☐ It is feasible in my setting (patients, time, data).
- ☐ The inclusion criteria describe the P (Population) of my PICO and can be checked from records on day 1.
- ☐ Every exclusion criterion has a reason (safety, outcome cannot be measured, and so on).
- ☐ I know what I will be allowed to claim: "caused" or "associated with".